Mt. St. HENRY'S HIGH SCHOOL MUKONO

ASPIRATION FOR EXCELLENCE

S.4 PHYSICS SEMINAR HELD ON SATURDAY 28TH, JUNE 2026

THE SEMINAR ITEMS BOOKLET

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WE WISH YOU SUCCESS IN YOUR NATIONAL EXAMINATIONS MOST ESPECIALLY PHYSICS SUBJECT

ITEMS PER ELEMENT OF CONSTRUCT

LIGTH AND WAVES

Item 1 (MBALALA S.S)

A boy taking cows to graze on a hot day observed that the air above the short grass appeared like colourless flames of fire which surprised him but he had no body to explain it. As the cows were grazing at the water dam reservoir surrounded by high hills, he threw his grazing stick towards one of the cows but he missed it and instead the stick dropped in the water with part of it out of the water.

When he went to pick the stick, it appeared broken which left him surprised again because water was too soft to break a stick.

The boy shouted calling his friends who were far to come and see a rear occurrance but he could hear a second sound 1.5 seconds later which appeared like some body was imitating him. The boy has come to you for assistance.

Hint: speed of sound in still air = 320 ms^{-1}

Task

Help him to understand

a) the occurances along the way to the water reservoir and at the water reservoir.
b) the cause of the second sound.
c) how far the hills are from the water reservoir
d) the applications of the concept that caused the occurance a long the way to the water reservoir.

Item 2

A newly established entertainment centre in a busy trading centre began organizing outdoor music and movie nights to attract customers. To make the events more exciting, the owner installed powerful loudspeakers around the premises and colourful stage lighting for performers.

The owner assured the community that the sound system was harmless because the sound produced had a wavelength of only 2.0 cm. However, residents living near the entertainment centre, particularly elderly people and learners preparing for examinations, complained that the sound was more disturbing at night than during the day.

During one of the events, performers wore costumes decorated with yellow, black and red stripes. As blue and green stage lights flashed across the stage, many spectators noticed that the colours of the costumes appeared to change. Some spectators claimed that the lights were actually changing the colours of the materials.

Concerned about the complaints and the differing opinions among community members, the local council invited you to explain the observations scientifically.

Hint: The speed of sound in air may be taken as 330 ms^{-1}

Task

As a physics learner;

a) Explain why sound from the entertainment centre is heard more clearly and causes greater disturbance to nearby residents at night than during the day.
b) evaluate the owner's claim that the sound is harmless.
c) Explain why the performers' costumes appeared to change colour under the flashing blue and green lights.
d) advise the entertainment centre owner and the local council on measures that can be taken to reduce noise disturbance while maintaining a safe and enjoyable environment for customers.

ATOMIC MODELS AND NUCLEAR PROCESSES

Item 3 (MT.ST. HENRY'S HIGH SCHOOL – MUKONO)

A patient diagnosed with an organ disorder at the Cancer Institute was treated using Iodine-131 and isolated in a specialized room where only well-protected workers could gain access. The patient's relatives were worried because they did not know how long their patient would stay in that room and why he was isolated.

After consulting the doctors, they were informed that the substance used is radioactive, which decays spontaneously with the emission of radiation and energy according to the following equation:

$$^{131}_{53}I \rightarrow {}^{131}_{54}X + Radiation + Energy$$

The doctors further explained to the relatives that the isolation room is equipped with an automatic system that records the count rate over time. The patient will be set free when the count rate drops to the background count rate of 50 counts per minute. The recorded results are shown below:

Time (days)	Activity (counts/min)
0	850
2	630
4	450
6	335
8	250
10	195
12	150

The relatives have contacted you for assistance.

Task

Help them to understand

a) Why the patient should not be given freedom to go everywhere.
b) The nature and properties of the radiation emitted in the given equation.
c) When the patient is likely to be set free (based on the provided data).
d) three applications of substances with properties similar to Iodine-131 in real life.

Item 4

A group of Senior Four learners visited a nuclear energy station as part of their career guidance and science learning programme. Upon arrival, they were informed that access to the facility had been temporarily suspended following an incident in one section of the plant.

The learners observed emergency response teams transporting some employees to a medical centre for examination using imaging equipment. Plant officials explained that radioactive materials had been released within a restricted area and that monitoring was being carried out continuously to determine when normal operations could safely resume.

Many learners became concerned and started discussing the situation. Some believed that all forms of radiation are equally dangerous, while others wondered how a power station could generate electricity from radioactive materials. A few learners also questioned why they had been denied entry when the station supplied electricity to a large part of the region.

The monitoring unit at the plant recorded the following information:

- Radiation level with of the source present = 520 counts per minute
- Maximum safe level for re-entry = 95 counts per minute
- Half-life of the radioactive material = 25 minutes

The plant manager has requested your assistance in helping the learners understand the situation scientifically.

Task

As a physics learner;

a) determine how long it will take before the radiation level falls to a value that allows the affected section of the plant to be safely used again. Hence advise the plant manager on when access may be restored.

b) help the learners understand:

i. The nuclear process responsible for the production of energy in a nuclear power station.

ii. How the energy released is converted into electrical energy for distribution to consumers.

c) describe the main types of radiation emitted by radioactive substances and compare their properties in terms of:

- Penetrating power
- Ionising ability
- Behaviour in electric and magnetic fields

d) With an illustration explain how X-rays used in medical diagnosis are produced.

e) justify the decision to deny the learners access to the facility and explain how workers can safely operate within a nuclear power station.

EARTH, SPACE PHYSICS AND DIGITAL ELECTRONICS.

Item 5

Learners in a certain school formed a space and technology club and one of the major club activities is to study Space, make observations, and look for reasons for the different observations made.

The club members have the following pending observations made recently without explanation:

- The moon has bright and dark sections and it changes its phase on a daily basis.
- The Lunar calendars are slightly different from the Gregorian (Solar) Calendars.
- The sky is full of stars but of different colours, brightness, and many of them twinkle.

The models made by the learners in their club activities are kept in a room where an automatic fire alarm was installed. The circuit consists of a logic gate where the alarm sounds in case of any smoke in the room and high temperatures, but many learners want to know how the system works.
You have been contacted for assistance.

Task

Help the learners to understand:

a) The appearance of the moon.
b) What causes the difference in the calendars.
c) Why the stars appear as observed.
d) The type of logic gate used, including a truth table.

Item 6

During lunch break, learners at a certain secondary school gathered around a tablet computer to watch a live international news broadcast. The report showed a country experiencing severe snowstorms, freezing temperatures and darkness, while people were being advised to remain indoors. The learners were surprised because at their school it was a bright, warm afternoon and outdoor activities were continuing normally.
Some learners argued that the country was cold because it was farther away from the Sun, while others believed that the Sun shines only on Africa during the day. Another learner wondered how the pictures and sounds from such a distant country were reaching their school instantly.
These learners have come to you to help them with explanation.

Task

Help the learners to understand

a) why one place on Earth can experience warm sunny conditions while another place experiences snowstorms and extremely low temperatures at the same time.
b) why it was daytime at the school while it was nighttime in the country experiencing the snowstorm.
c) The process through which information from the distant country reaches the tablet in real time.
d) Other applications of applications of the signals used in the process described above.

HEAT AND MECHANICS

ITEM 7 (MPOMA SCHOOL SATELLITE CAMPUS)

In a bid to beef up security at the school, a temporary gate is to be erected by using a uniform metallic beam 5m long and of weight 100N pivoted at a point 50cm from one end and a metallic

chain of mass 2kg at its extreme free end. To balance the metallic beam horizontally, the Askari has been advised to fix a bag of rock dust from its end near the pivot.
The Askari found out that rock dust is sold at shs.500 per kilogram and he has requested and given shs.25000 to buy the rock dust but he is not sure whether the money is enough to buy the required rock dust.
At break, the Askari was served with hot tea in a black metallic cup of volume 0.4 litres at a temperature of 80°C and immediately, he added 100g of sugar and stirred using a metallic spoon which made the temperature to reduce to 60°C and after 30 minutes, the temperature of tea reduced further to 45°C which surprised the Askari.
The Askari has contacted you for assistance.

Hint

* Specific heat capacity of tea = 4000J kg^{-1}K^{-1}
* Density of tea = 1200kg m^{-3}
* Acceleration due to gravity, g = 10ms^{-2}

Task.

Help the Askari to understand:

a) whether the money given to Askari will be enough to obtain the required Rock dust
b) the specific heat capacity of Sugar stating any assumptions made.
c) the temperature of the tea after 40 minutes.
d) reasons why the temperature of the tea was reducing at such a high rate.
e) the applications of blackened surfaces in our everyday life

Item 8

A timber factory has been constructed on a 40 m high hill in a remote village where piped water is not available. The factory uses a large amount of water to cool the hot moving parts of the saw mill to prevent overheating during operation.

A driver delivers water to the factory using a 5000 kg truck fitted with a water tank of 3000 litres when full. To earn more money per trip, the driver decided to carry half a tank of water.

During operation of the saw mill, the cooling water enters the machine at 20°C and leaves at 70°C after absorbing heat from the hot moving parts. The factory engineer estimates that only 80% of the heat produced by the mill is transferred to the cooling water, while the remaining heat is lost to the surroundings.

One day, while delivering water to the factory, the driver accelerated the truck uniformly from rest to a speed of 90 km h^{-1} in 5 minutes, maintained this speed for 10 minutes, and then suddenly applied the brakes, bringing the truck to rest instantly. He observed a lot of smoke between the tyres and the road surface. At the same time, the truck shook vigorously **to** and **fro** for a short time, almost causing an accident. The driver could not explain these observations and has contacted you for assistance.

Hint

- Specific heat capacity of water = 4200 J kg^{-1} K^{-1}
- Density of water = 1000 kg m^{-3}

- Acceleration due to gravity, g = 10 m s^{-2}

Task

Help the driver to:

a) determine the gravitational potential energy gained by the truck and the water after reaching the top of the hill.
b) represent the motion of the truck using a suitable graph and hence determine its average velocity.
c) explain the cause of the smoke observed between the tyres and the road surface, and the vigorous shaking of the truck after it stopped.
d) determine:

   (i) the heat absorbed by the cooling water while passing through the saw mill, and

   (ii) the total heat produced by the saw mill.

ITEM 9

Two mountain climbers A and B have been given a job to take meat to tourists who have camped at the top of the mountain. They use a uniform wooden pole of length 2m and weight 50N to carry the meat. On a particular day they had to supply 50kg of meat to the team and it was tied at 50cm mark from one end of the pole. The climbers felt great tension at their shoulders as they carried it and they wondered why the pole didn't break despite the heavy load which was being carried using it for some time. When they reached home, the team had sent a complaint that they delayed the supply and they had to deliver the meat in the next 30minutes. However, they were hungry and one of then urged that 10 minutes are enough to move from home to the supplier, so they agreed to first prepare part of the meat using an electric air-frier rated 750W and eat. If the heat capacity of meat being cooked is 16.5KJKg^{-3} and the meat will get ready if its temperature raises by 120℃

On the next day they become curious and wanted to know the height of the mountain so they used a barometer to estimate the atmospheric pressure at the bottom and the top which were recorded has 76 cmHg and 72.12 cmHg respectively.

One of the climbers complained that he was nose bleeding has they continued to climb the mountain and was left in wonders.

They have decided to approach you for help

Task:

As a physics learner:

a) Determine the forces that caused pain at their shoulders.
b) Establish whether the climbers will be able to deliver the meat in time if they consider eating first.
c) Estimate the height of the mountain if the density of mercury is 13600kg$m^{-3}$and air is 1.2kgm^{-3}.

d) Explain to the climbers why the complaint of one of them is valid.
e) Help the climber know why the wooden pole did not break.

ELECTRICITY AND MAGNETISM

Item 10

Kato bought two sets of solar systems one for himself and another for his parents in the village. He asked a technician to install one on his house and fixed a device that records the EMF of the battery as it charges and as it discharges with time. After a few days, Kato noted that the bulbs become dim and eventually go off when he is sleeping.

One day he recorded data from the recording device but failed to make use of it.

Emf(V)	24.0	20.0	16.0	12.0	8.0
Time (minutes)	0	60	120	220	300

Kato knew that he had acquired enough skills and knowledge to install the system on the house of the parents and it failed to work which left him wondering, because it could only work during the day.

He also bought an electric kettle whose label was scratched and only its voltage of 240V and coil resistance of 12 Ω could be seen yet he wanted to know the power and current that flows through the kettle.

Hint:

- ✓ The bulbs go off when the battery voltage goes below 10.0V.
- ✓ The system stops charging when the emf goes below 6.0V.

You have been contacted by Kato for assistance.

Task

Help Kato to understand:

a) the time when the bulbs go off if he switches them on at 7:00pm **using a graph.**
b) the circuit he can use with reasons for the system on the parents' house to work well.
c) the power of the kettle and the current it uses.
d) other two uses of electricity other than heating and lighting in our everyday life.

Item 11

When you visited your uncle in one of the developing towns in Uganda, you found when the authorities had ordered for the breaking and removal of all kiosks in a bid to clean and organize the town better.

The process left many sharp rusty metals and pieces of glass that have proved hard to pick and your uncle is wondering the method to use to clean the place.

Of recent the D.C motor of your uncle's grinding mill heats up so much and produces a lot of noise which is worrying your uncle. He does not know how it works and how its efficiency can be improved.

The uncle also realized that when 240V and current of 30A flows for 10 minutes in the motor, it does work of 2.7×10^6 J.

Your uncle has contacted you for help.

Task.
Help your uncle to understand:

a) the design of the device that can be used to pick the metals and why it may not be suitable for removing the pieces of glass.
b) how the motor is able to spin the milling machine in one direction.
c) (i) the cause of the noise and how this noise can be reduced.
(ii) the efficiency of the motor
d) where the motors are used other than milling.

Item 12

A homeowner has a house connected to a **240 V mains supply**. In his home he has a radio set labeled **120 V, 90 W,** An electric kettle labeled **120 V, 2400 W, a**n electric device with a resistance of **8 Ω** installed at the gate to alert occupants when visitors press a switch and he has always wondered how sound gets to be produced in the process, **6 lamps** each labeled **120 V, 100 W**, to be used for lighting kept on 5 hours day and a power extension fitted with a **10 A fuse**, into which the radio and electric kettle are plugged. The radio is used for **8 hours per day**. The electricity tariff charged by the power company is **UGX 750 per unit.** He has some unanswered questions about the usage of electricity in his home and needs to consult someone with technical knowledge, you have paid him a visit and he is lad to receive you and is aware that you're in senior four.

Tasks

As a learner of physics, assist him in the following areas,

a) Comment on whether the **10A fuse** is suitable for protecting the extension when both appliances are operating simultaneously.
b) State and explain to him the most appropriate way to connect the six lamps in the house.
c) Explain to him how the electric device produces sound when the switch at the gate is pressed.
d) Determine the daily cost of operating the radio and the six lamps.

EXTRA PRACTICE ITEM.

Item 13

After visiting China, your father bought a D.C motor rated 80V, 12000W and a generator which can produce alternating voltage of 240V with the intention of starting a home factory for processing ground nut and simsim paste for value addition. He wonders how these two these two devices can be connected given the fact he was not briefed while in China. He is suspecting that his products may contain many metallic particles due to wear and tear which is disturbing him.

Task

As a physics learner,

a) draw a clear set up including suitable devices showing how the two devices can be connected.
b) Explain to him on how to clean his products.
c) In three hours, 10kg of the mixture can be milled and unit costs shs.1000. explain to him the production cost.

Item 14

A local entrepreneur is setting up a small woodworking workshop and is to have 4 LED ceiling lights (each labeled 10 W, 220 V), 2 electric drills (each labeled 500 W, 220 V), 1 electric saw (labeled 1200 W, 220 V), 1 electrical lifting device used to pick up metal tools (labeled 220 V, 2 A), he wants the lifting device to be very strong as he plans to lift heavy metals. The workshop is connected to a standard commercial electricity supply of 220 V and he has acquired a 15A circuit breaker. The entrepreneur needs assistance on how to design a safe and efficient wiring system.

Task:

As a learner of physics, assist the entrepreneur to;

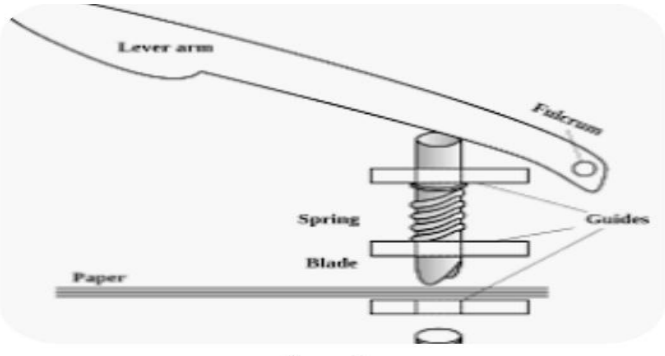
a) Know with reason the best way to connect the appliances.
b) Determine whether a 15A circuit breaker would be sufficient.
c) Understand how the lifting device works and suggest one way to increase the strength without changing the voltage supply.
d) Identify at least two safety features that should be included in the workshop wiring.

PRACTICAL ITEMS

ITEM ONE (1):

One afternoon, the school secretary was drilling holes on the papers for filing using a heavy duty punching machine, but suddenly the punching machine stopped working. The mechanic discovered that the spring of force constant between **$200Nm^{-1}$** and **$600Nm^{-1}$**, in the punching machine was damaged and required replacement.

The secretary bought a new spring from a nearby hardware, but without specification and sought for help from you.



TASK ONE (1):

You are provided with a spring from the laboratory, with identical properties as the one bought by the secretary and whose force constant is **twenty times** smaller than that in the punching machine, advice on its suitability for use in the punching machine.

HINT: Include the following in your report;

Aim, variables (independent, dependent and controlled), hypothesis, procedure, errors anticipated, precautions to mitigate the errors, present data accurately (6 sets), analyse data (using graph) and advice/recommend.

ITEM TWO (2):

A builder fixing windows for a community member proposed that the best glass for the window shutter should be of critical angle in the range of **38º** to **42º** inclusive. The community member moved to a nearby shop that sells glasses and bought a large sheet of glass but was not sure of its critical angle and contacted you for assistance.



TASK TWO (2):

You are provided with a glass block from the laboratory, with similar properties as the large sheet of glass bought by the community member, advise accordingly on its fitness for window shutter.

HINT: Include the following in your report;

Aim, variables (independent, dependent and controlled), hypothesis, procedure, errors anticipated, precautions to mitigate the errors, present data accurately (6 sets), analyse data (using graph) and advice/recommend.

ITEM THREE (3):

Some of the community members complain about the new dry cells being sold in the shops taking just a short time only and getting used up. One member suggested that those dry cells had higher internal resistance and that a standard dry cell has an internal resistance in the range of 0.1Ω and 0.6Ω inclusive. One of the community members bought a new pair of dry cell from the shop but could not ascertain its internal resistance and contacted you for assistance.



TASK THREE (3):

You are provided with one dry cell from the laboratory, with same properties as the new pair the community member bought from the shop, advice accordingly.

HINT: Include the following in your report;

Aim, variables (independent, dependent and controlled), hypothesis, procedure, errors anticipated, precautions to mitigate the errors, present data accurately (6 sets), analyse data (using graph) and advice/recommend.

EXTRA PRACTICAL ITEM

A laboratory technician wishes to store 150 identical glass marbles in a jar that can safely support a maximum mass of 4 kg. However, the mass of each marble is unknown, and the weighing scale in the laboratory is faulty.

The technician has therefore requested your assistance in determining whether the jar can safely accommodate all the marbles.

Task:

You have been given one of the marbles, use it to ascertain whether the 150 identical marbles can fit in the container.

END